



Figure 2: **Circuit schemes for the finite differences and local circuit perturbation approaches.** Circuit to compute the RGF as in Eq. (8) between the first and last site with a single (local-time) circuit perturbation $R^\beta(\cdot)$. The argument of the rotation gate is a small positive angle $\pm\epsilon/2$ for the FD approach (left) and $\pm\pi/2$ in the LCP approach (right). $|\psi_0\rangle$ is the initial state, T the total simulation time, N the number of Trotter steps and $\bar{\tau} = \tau(\bar{n} - 1)$, $\tau = T/N$ (see also main text). $U(\cdot)$ indicates the time evolution operator, $U(0) \equiv \mathbb{1}^{\otimes L}$ and the quantum register is composed of L qubits.

where

$$\mathcal{G}(\mathbf{R}, \mathbf{r}', T, t') = -i\Theta(T - t') \langle [o_{\mathbf{R}}^I(T), o_{\mathbf{r}'}^I(t')] \rangle \quad (7)$$

is a generalized susceptibility, with the step function $\Theta(T - t')$ explicitly enforcing causality. This object can be directly identified with the RGFs of the observable, an essential quantity that characterizes the system's causal response to external perturbations and provides key information about its dynamical and spectral behavior. Interestingly, Eq. (6) also shows that the latter can be expressed as the functional derivative

$$\mathcal{G}(\mathbf{R}, \mathbf{r}', T, t') = \left. \frac{\delta \langle o_{\mathbf{R}}(T) \rangle_s}{\delta s(\mathbf{r}', t')} \right|_{s=0}. \quad (8)$$

As shown in the next section, this formulation provides the basis for the development of several quantum algorithms to compute the RGF.

3 Methods

The relation in Eq. (8) enables the definition of a class of estimators for RGFs based on circuit derivatives. In this section, we illustrate this framework through concrete examples showing how the required functional derivatives can be accessed using (one or more) parametrized quantum circuits. The working assumption is that a quantum register can be initialized in the ground state $|\psi_0\rangle$ and that both the unperturbed and perturbed propagators can be efficiently implemented – e.g., via a product formula – in a digital quantum simulation. Moving back to the Schrödinger picture, we also assume that – in the spirit of a Suzuki-Trotter decomposition – the action of H_0 and $H_s(t')$ can be factorized to yield a total propagator up to time T of the form

$$U_s(T, N) \approx \prod_{n=0}^{N-1} \left[e^{-iH_0\tau} e^{-iH_s(n\tau)\tau} \right], \quad (9)$$

where $\tau = T/N$ and H_s is evaluated on a discrete set of time points. We can then construct the functional

$$\mathcal{F}[s] := \langle U_s^\dagger(T, N) o_{\mathbf{R}} U_s(T, N) \rangle, \quad (10)$$

which corresponds to the digital quantum simulation of the expectation value $\langle o_{\mathbf{R}}(T) \rangle$ with a specific choice of the force function $s(\mathbf{r}', t')$ across positions $\mathbf{r}'$ and times $t' \in [0, T]$ for a total evolution time T .

3.1 Local circuit perturbation

By using a Dirac delta in both position and time as a test function, the functional derivative of $\mathcal{F}$ with respect to s can formally be expressed as

$$\frac{\delta \mathcal{F}[s]}{\delta s} = \lim_{\epsilon \rightarrow 0} \frac{\mathcal{F}[s + \epsilon \delta(\mathbf{r}' - \mathbf{r}) \delta(t' - t)] - \mathcal{F}[s]}{\epsilon}. \quad (11)$$

Recalling from Eq. (8) that $\mathcal{G}$ is given by the above expression evaluated around a constant null perturbation $s = 0$, we can approximate it via the Finite Differences (FD) expression

$$\mathcal{G}(\mathbf{R}, \mathbf{r}, T, t) \approx \left[\langle U_{\delta s}^\dagger(T, N) o_{\mathbf{R}} U_{\delta s}(T, N) \rangle - \langle U_0^\dagger(T, N) o_{\mathbf{R}} U_0(T, N) \rangle \right] \epsilon^{-1}, \quad (12)$$

where $\delta s = \epsilon \delta(\mathbf{r}' - \mathbf{r}) \delta(t' - t)$; with some abuse of notation, we identify from here on $U_{\delta s} \equiv U_\epsilon$. The time evolution in the first line of Eq. (12) contains a contribution generated by the space- and time-local perturbation

$$H_\epsilon(t'; \mathbf{r}, t) = \epsilon \delta(t' - t) o_{\mathbf{r}}, \quad (13)$$

with $\epsilon \ll 1$, i.e., a small kick applied to the system at position $\mathbf{r}$ and time t , whose parametric dependency is indicated after the semicolon. The second term of Eq. (12) is instead the ground-state expectation value $\langle o_{\mathbf{R}} \rangle$, which remains constant under the unperturbed evolution assuming negligible Suzuki-Trotter discretization errors. Notice that, while in Eq. (11) we used the so-called forward definition of the FD ratio, the accuracy can be improved from $\mathcal{O}(\epsilon)$ to $\mathcal{O}(\epsilon^2)$ by using a symmetric increment of $\pm\epsilon/2$. Such higher-order version (see Fig. 2) will be adopted henceforth.

To be more concrete, assume that $o_{\mathbf{R}} = \sigma_R^\alpha$ and $o_{\mathbf{r}} = \sigma_r^\beta$ are, for instance, single qubit Pauli operators acting on a one-dimensional spin-1/2 lattice. In a quantum circuit representation, the first term on the RHS of Eq. (12) is then obtained by introducing a